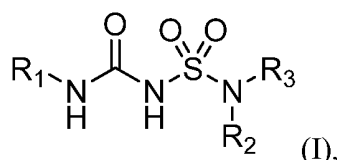


## CLAIMS:

1. A compound of Formula (I):



or a solvate, or pharmaceutically acceptable salt thereof, wherein:

**R<sub>1</sub>** is C<sub>3</sub>-C<sub>16</sub> cycloalkyl, or C<sub>5</sub>-C<sub>10</sub> aryl, wherein the C<sub>3</sub>-C<sub>8</sub> monocyclic cycloalkyl, polycyclic cycloalkyl, or C<sub>5</sub>-C<sub>6</sub> aryl is optionally substituted by one or more **R<sub>1S</sub>**; wherein each **R<sub>1S</sub>** is independently C<sub>1</sub>-C<sub>6</sub> alkyl, C<sub>1</sub>-C<sub>6</sub> haloalkyl, C<sub>1</sub>-C<sub>6</sub> alkoxy, C<sub>1</sub>-C<sub>6</sub> haloalkoxy, or halo;

**R<sub>2</sub>** is -(C**X<sub>2</sub>****X<sub>2</sub>**)<sub>n</sub>-**R<sub>2S</sub>**, wherein n is 0, 1, or 2, and each **X<sub>2</sub>** is independently H, C<sub>1</sub>-C<sub>6</sub> alkyl, C<sub>2</sub>-C<sub>6</sub> alkenyl, or C<sub>2</sub>-C<sub>6</sub> alkynyl, wherein the C<sub>1</sub>-C<sub>6</sub> alkyl, C<sub>2</sub>-C<sub>6</sub> alkenyl, or C<sub>2</sub>-C<sub>6</sub> alkynyl is optionally substituted with one or more halo, -CN, -OH, -O(C<sub>1</sub>-C<sub>6</sub> alkyl), -NH<sub>2</sub>, -NH(C<sub>1</sub>-C<sub>6</sub> alkyl), -N(C<sub>1</sub>-C<sub>6</sub> alkyl)<sub>2</sub>, or oxo;

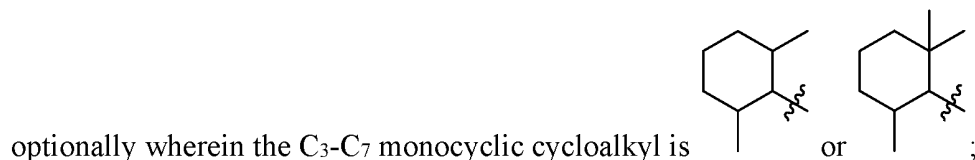
**R<sub>2S</sub>** is 4- to 8-membered heterocycloalkyl optionally substituted with one or more C<sub>1</sub>-C<sub>6</sub> alkyl, C<sub>2</sub>-C<sub>6</sub> alkenyl, C<sub>2</sub>-C<sub>6</sub> alkynyl, C<sub>1</sub>-C<sub>6</sub> haloalkyl, halo, -CN, -OH, -O(C<sub>1</sub>-C<sub>6</sub> alkyl), -NH<sub>2</sub>, -NH(C<sub>1</sub>-C<sub>6</sub> alkyl), -N(C<sub>1</sub>-C<sub>6</sub> alkyl)<sub>2</sub>, or oxo; and

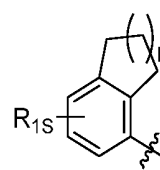
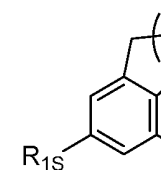
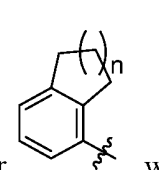
**R<sub>3</sub>** is 7- to 12-membered heterocycloalkyl or 5- or 6-membered heteroaryl optionally substituted with one or more **R<sub>3S</sub>**, wherein each **R<sub>3S</sub>** is independently C<sub>1</sub>-C<sub>6</sub> alkyl, C<sub>1</sub>-C<sub>6</sub> haloalkyl, C<sub>3</sub>-C<sub>8</sub> cycloalkyl, halo, or C<sub>3</sub>-C<sub>8</sub> heterocycloalkyl wherein the C<sub>1</sub>-C<sub>6</sub> alkyl, C<sub>1</sub>-C<sub>6</sub> haloalkyl, C<sub>3</sub>-C<sub>8</sub> cycloalkyl, or C<sub>3</sub>-C<sub>8</sub> heterocycloalkyl is optionally substituted with -O(C<sub>1</sub>-C<sub>6</sub> alkyl), -N(C<sub>1</sub>-C<sub>6</sub> alkyl)<sub>2</sub>, halo, or -CN.

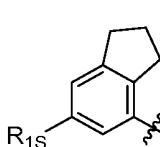
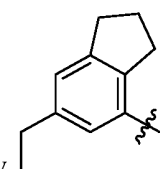
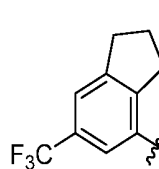
2. The compound of any one of the preceding claims, wherein **R<sub>1</sub>** is C<sub>3</sub>-C<sub>7</sub> monocyclic cycloalkyl, C<sub>9</sub>-C<sub>10</sub> bicyclic cycloalkyl, C<sub>12</sub>-C<sub>16</sub> tricyclic cycloalkyl, or C<sub>5</sub>-C<sub>10</sub> aryl, wherein the C<sub>3</sub>-C<sub>7</sub> monocyclic cycloalkyl, C<sub>9</sub>-C<sub>10</sub> bicyclic cycloalkyl, C<sub>12</sub>-C<sub>16</sub> tricyclic cycloalkyl, or C<sub>5</sub>-C<sub>10</sub> aryl, is optionally substituted by one or more **R<sub>1S</sub>**;

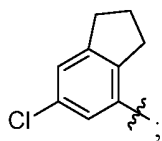
optionally wherein:

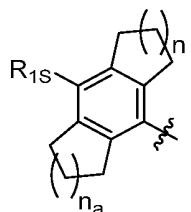
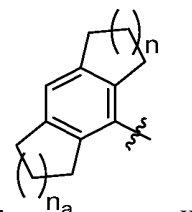
(a) the C<sub>3</sub>-C<sub>7</sub> monocyclic cycloalkyl is cyclopentyl, cyclohexyl, or cycloheptyl, wherein the cyclopentyl, cyclohexyl, or cycloheptyl is optionally substituted by one or more **R**<sub>1S</sub>; further

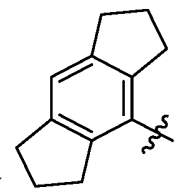


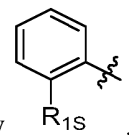
(b) the C<sub>9</sub>-C<sub>10</sub> bicyclic cycloalkyl is , , or , wherein n

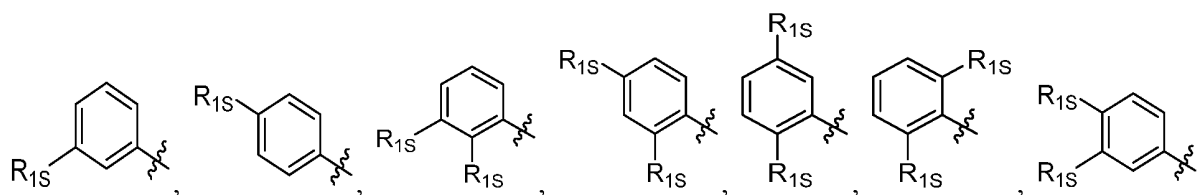
is 0, 1, or 2; optionally ; further optionally , , or

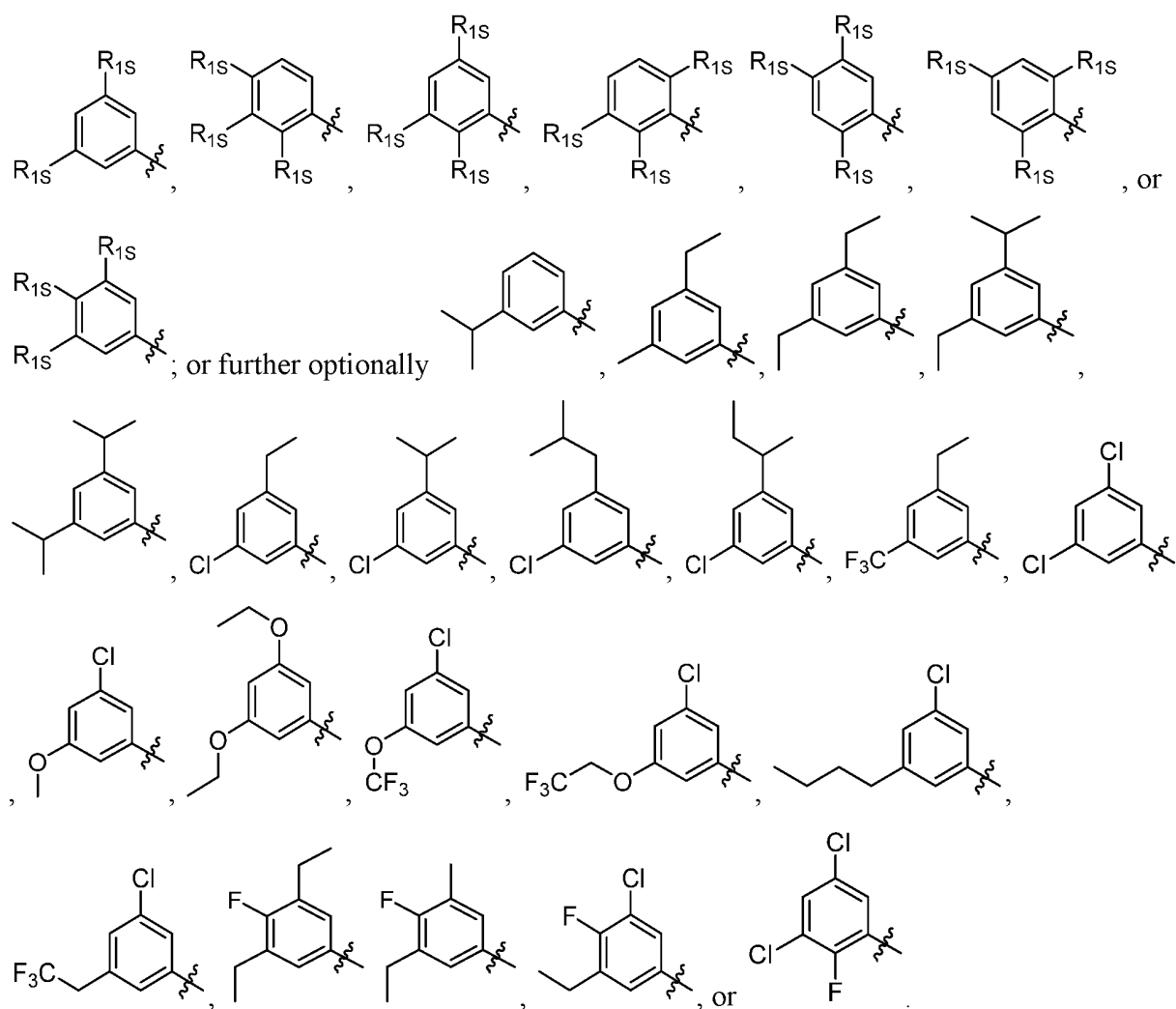


(c) the C<sub>12</sub>-C<sub>16</sub> tricyclic cycloalkyl is , or , wherein n and n<sub>a</sub>

each independently are 0, 1, 2, or 3; further optionally ; and/or

(d) the C<sub>5</sub>-C<sub>10</sub> aryl is phenyl is substituted by one, two, or three **R**<sub>1S</sub>; optionally ,



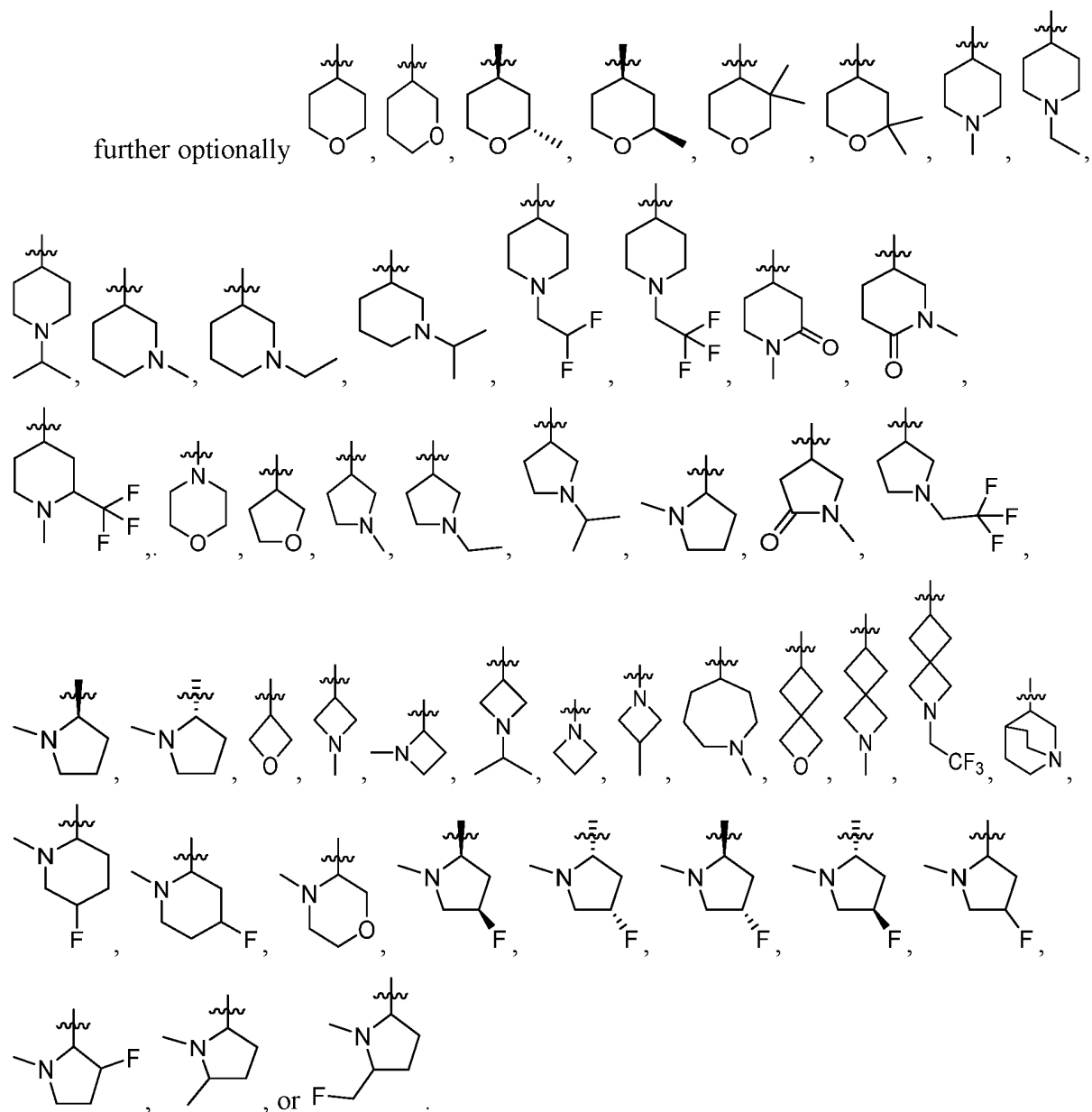


3. The compound of claim 1 or 2, wherein **R<sub>2</sub>** is -**R<sub>2S</sub>**, -(**CX<sub>2</sub>X<sub>2</sub>**)-**R<sub>2S</sub>**, or -(**CX<sub>2</sub>X<sub>2</sub>**)<sub>2</sub>-**R<sub>2S</sub>**.

4. The compound of any one of claims 1 to 3, wherein **R<sub>2S</sub>** is 4- to 8-membered heterocycloalkyl optionally substituted with one or more C<sub>1</sub>-C<sub>6</sub> alkyl, C<sub>2</sub>-C<sub>6</sub> alkenyl, C<sub>2</sub>-C<sub>6</sub> alkynyl, C<sub>1</sub>-C<sub>6</sub> haloalkyl, halo, -CN, -OH, -O(C<sub>1</sub>-C<sub>6</sub> alkyl), -NH<sub>2</sub>, -NH(C<sub>1</sub>-C<sub>6</sub> alkyl), -N(C<sub>1</sub>-C<sub>6</sub> alkyl)<sub>2</sub>, or oxo;

optionally oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, azepanyl, 2-azaspiro[3.3]heptane, 2-oxaspiro[3.3]heptane, or bridged quinuclidinyl, wherein the oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, azepanyl, 2-azaspiro[3.3]heptane, 2-

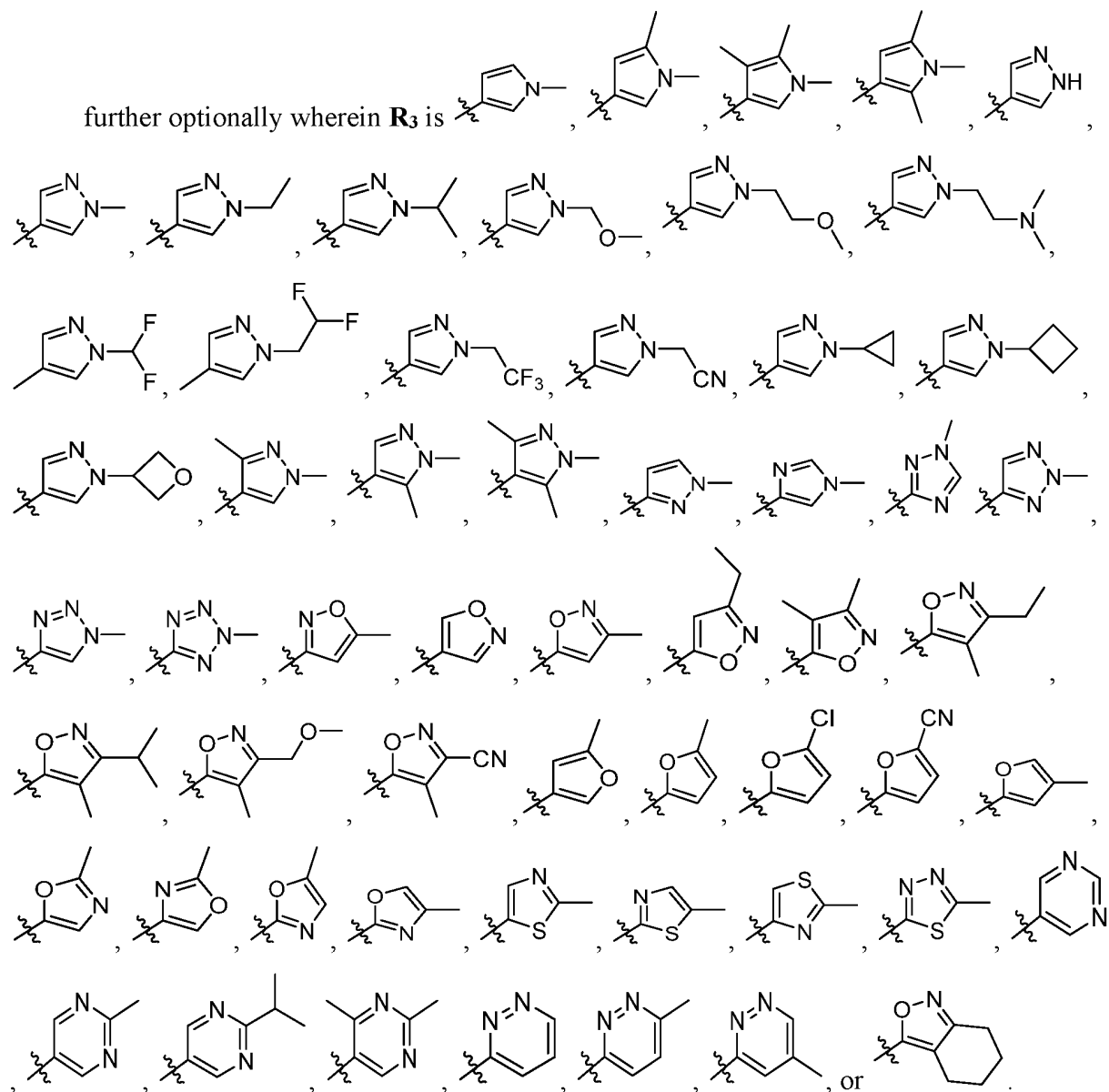
oxaspiro[3.3]heptane, or bridged quinuclidinyl is optionally substituted with one or more C<sub>1</sub>-C<sub>6</sub> alkyl, C<sub>1</sub>-C<sub>6</sub> haloalkyl, or oxo; or



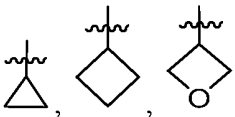
5. The compound of any one of claims 1 to 4, wherein **R**<sub>3</sub> is 7- to 12-membered heterocycloalkyl optionally substituted with one or more **R**<sub>3S</sub>;

optionally wherein **R**<sub>3</sub> is pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-tetrahydrobenzo[c]isoxazole, wherein the pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl,

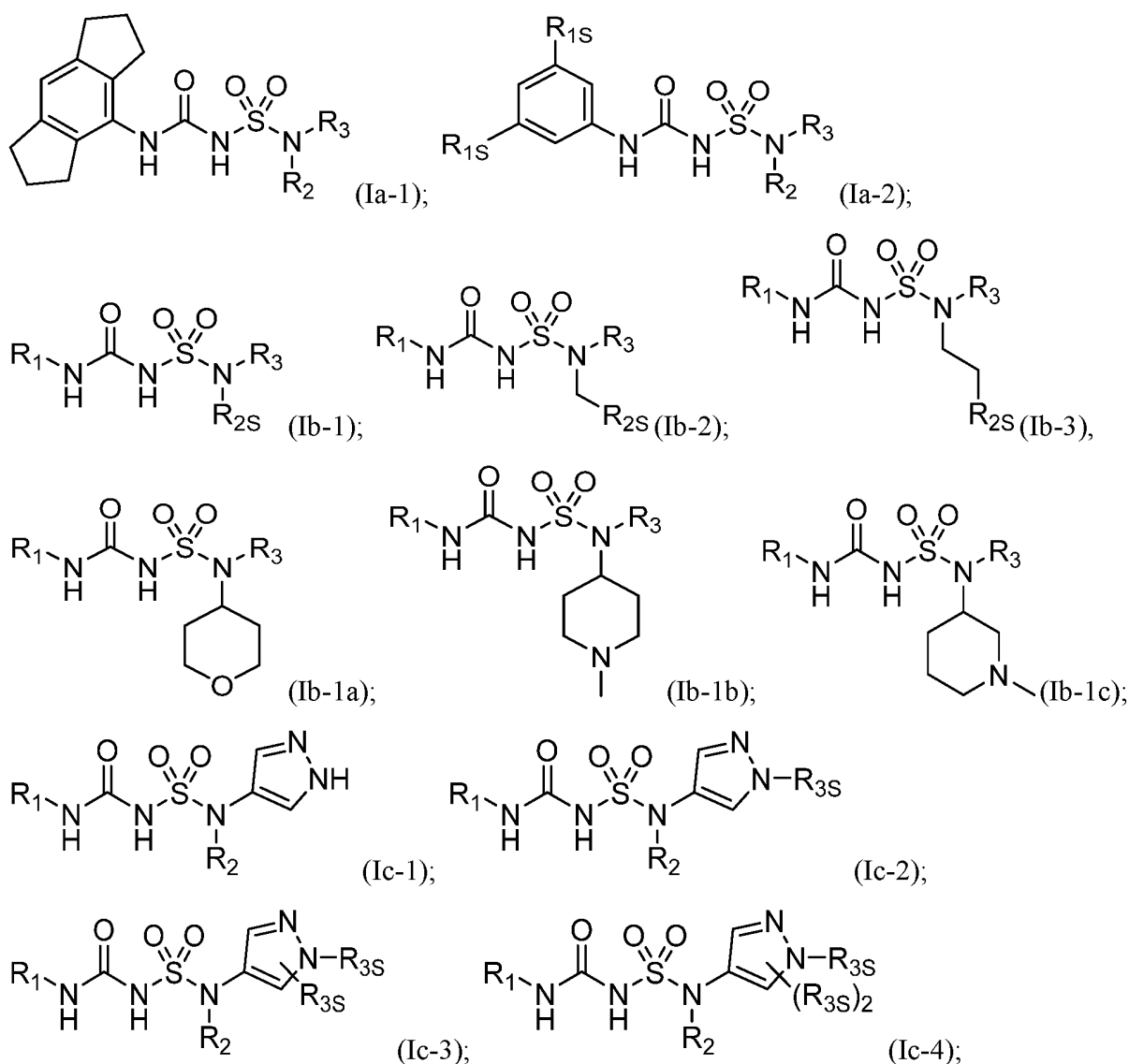
isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-tetrahydrobenzo[c]isoxazole is optionally substituted with one or more **R**<sub>35</sub>; or

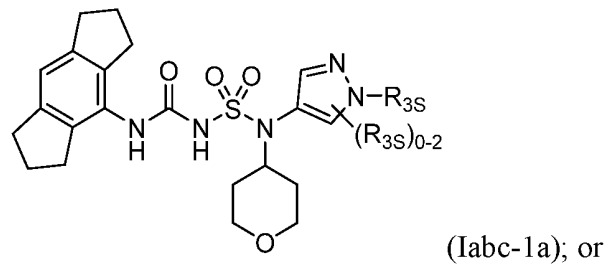
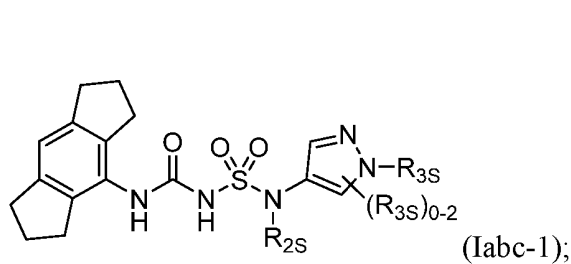
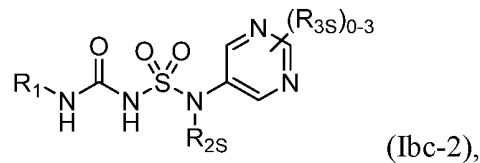
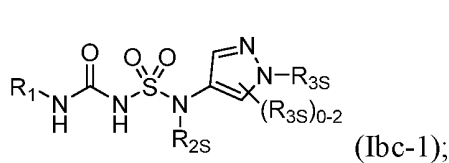
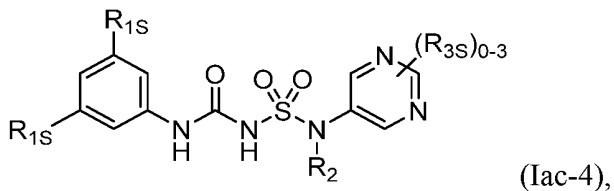
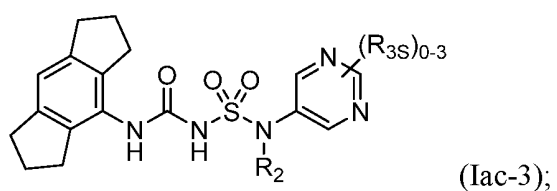
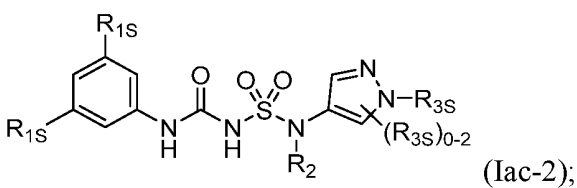
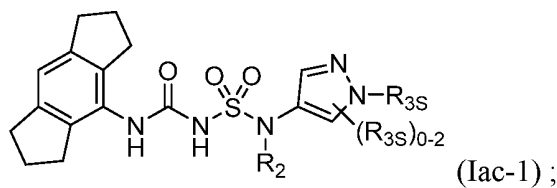
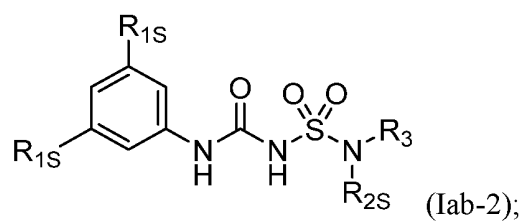
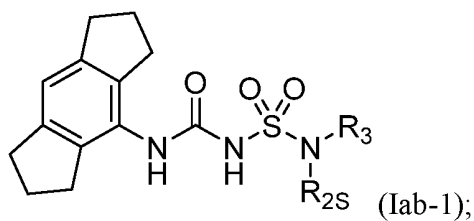
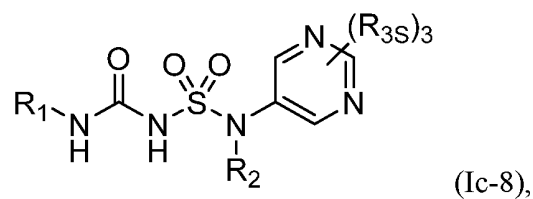
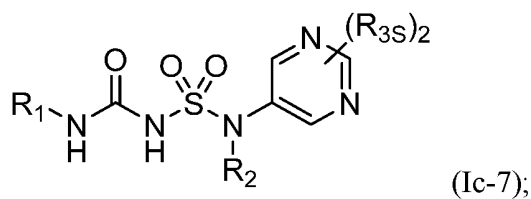
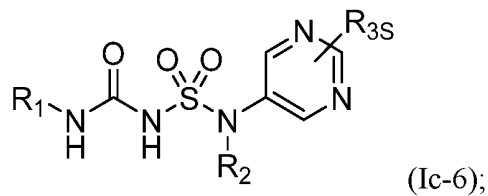
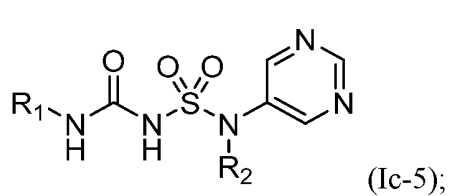


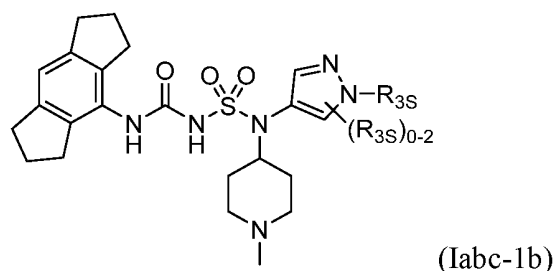
6. The compound of any one of claims 1 to 5, wherein at least one **R<sub>3S</sub>** is C<sub>1</sub>-C<sub>6</sub> alkyl, C<sub>1</sub>-C<sub>6</sub> haloalkyl, C<sub>3</sub>-C<sub>8</sub> cycloalkyl, halo, or C<sub>3</sub>-C<sub>8</sub> heterocycloalkyl wherein the C<sub>1</sub>-C<sub>6</sub> alkyl, C<sub>1</sub>-C<sub>6</sub> haloalkyl, C<sub>3</sub>-C<sub>8</sub> cycloalkyl, or C<sub>3</sub>-C<sub>8</sub> heterocycloalkyl is optionally substituted with -O(C<sub>1</sub>-C<sub>6</sub> alkyl), -N(C<sub>1</sub>-C<sub>6</sub> alkyl)<sub>2</sub>, halo, or -CN;

optionally wherein at least one  $R_{3S}$  is , methyl, ethyl, isopropyl, -CH<sub>2</sub>OCH<sub>3</sub>, -CH<sub>2</sub>CF<sub>3</sub>, -CH<sub>2</sub>CH<sub>2</sub>OCH<sub>3</sub>, -CH<sub>2</sub>CN, CH<sub>2</sub>CF<sub>2</sub>, or -CH<sub>2</sub>CH<sub>2</sub>N(CH<sub>3</sub>)<sub>2</sub>.

7. The compound of any one of the preceding claims, wherein the compound is of Formula (Ia-1), (Ia-2), (Ib-1), (Ib-2), (Ib-3), (Ib-1a), (Ib-1b), (Ib-1c), (Ic-1), (Ic-2), (Ic-3), (Ic-4), (Ic-5), (Ic-6), (Ic-7), (Ic-8), (Iab-1), (Iab-2), (Iac-1), (Iac-2), (Iac-3), (Iac-4), (Ibc-1), (Ibc-2), (Iabc-1), (Iabc-1a), or (Iabc-1b):

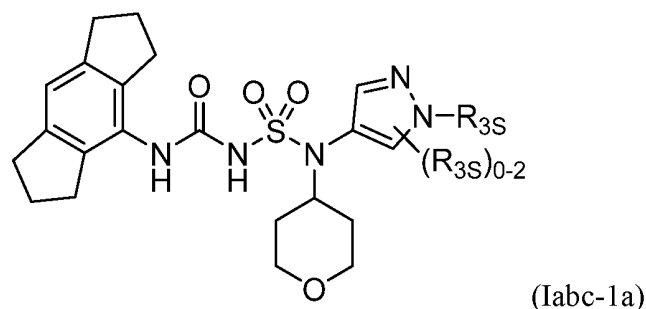






or a solvate, or pharmaceutically acceptable salt thereof, wherein **R<sub>1</sub>**, **R<sub>2S</sub>**, **R<sub>2</sub>**, and **R<sub>3</sub>** are as described herein.

8. The compound of any one of claims 1 to 7, wherein the compound is of Formula (Iabc-1a):



or a solvate, or pharmaceutically acceptable salt thereof, wherein **R<sub>3S</sub>** is as described herein.

9. The compound of any one of the claims 1 to 8, being selected from Compound Nos. 1-166 and pharmaceutically acceptable salts thereof.

10. A pharmaceutical composition comprising the compound of any one of claims 1 to 9 or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable diluent or carrier.

11. The compound of any one of claims 1 to 9, or the pharmaceutical composition of claim 10, for use in inhibiting inflammasome activity; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.

12. The compound of any one of claims 1 to 9, or the pharmaceutical composition of claim 10, for use in treating or preventing a disease or disorder; preferably wherein the disease or disorder

is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.

13. The compound or pharmaceutical composition for use of claim 11 or 12, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated autoinflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological diseases (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).

14. The compound or pharmaceutical composition for use of claims 11 or 12, wherein disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.

15. The compound or pharmaceutical composition for use of claims 11 or 12, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, gastrointestinal cancer, brain cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal adenocarcinoma.